

Mark Gasson, another British scientist, implanted an advanced glass capsule RFID device surgically into his left hand, in 2009. Later, he demonstrated how the implanted microchip can be perceived as being a part of the human body, showing that separation between man and machine can become theoretical. He also demonstrated how a computer virus wirelessly infected his implant and then got transmitted on to other systems.

Later, Amal Graafstra, an amateur biohacker, successfully used implants to access his home, open car doors and log on to his computer. He built a smart gun that could fire only after reading his implant.

Martijn Wismeijer, who was working as marketing manager for Bitcoin ATM manufacturer General Bytes, placed RFID chips in both his hands to store his private Bitcoin keys and business card.

Marcel Varallo inserted an NXP chip into his hand. He used it to open secure elevators and doors at work, print from secure printers, unlock his cellphone and home, and store his digital business card for transfer to cellphones enabled with near field communication (NFC) to make payments.

Technology behind human implant devices and apps

Recent developments in technology have made possible new potential applications for RFID devices implanted in the human body. These can now be used for remote identification of persons, remotely controlling human biological functions, transmitting measurements of chemical or biological data inside the body, and monitoring biological activity or physiological functions of humans.

A classic example is VeriChip tag, which contains a numerical code as identifier, readable from a distance of 10cm or less using a handheld reader.

When illuminated by a low-frequency magnetic field (134kHz) generated by the reader, the tag powers up and transmits a unique 16-digit code, which can be used for identifying individuals remotely. The technology is not yet fully developed for real-time tracking of people using an implanted



Fig. 3: VeriChip tag (Credit: <http://cq.cx/vchdiy.pl>)



Fig. 4: A remote control that allows patients to start, stop or pause therapy, and to charge sleep apnea implant (Credit: <http://imtheramedical.com>)

microchip. To be able to track anywhere on Earth, a GPS-controlled device would be required for this application. This would mean dimensions that are too large for implanting inside a human body.

Miniaturisation of such a device for building a rice-grain-sized GPS implant is a challenging technological issue that is yet to be solved. The current technology for human implants is based on battery-less (passive) devices, which allow achieving very short read range, typically 10cm or less.

Consequently, these implants cannot communicate with GPS satellites (or other equivalent satellite-based positioning systems) required for real-time tracking of people. Sub-micron-sized implants connected to GPS is not feasible using the technology available today.

Remote control of human biological functions. A single chip radio system with onboard antenna and sensors of size $100\mu\text{m} \times 100\mu\text{m} \times 1\mu\text{m}$ is what is feasible with current technology. Consequently, wireless communication becomes possible with human implants for receiving information about chemical or physical quantity in biological systems, or for provid-

ing remote activation/deactivation of biochemical activity inside a single living cell.

Using nanotechnology, fabrication of an FM radio using a single nano-tube of $1\mu\text{m}$ length and 10nm width is possible. The nano-tube radio is inserted inside the living human cell to give a sub-cellular remote and real-time controlled interface.

Miniaturisation of the radio chip could also be used in the fabrication of smart dusts, that is, tiny objects with sensing and communication abilities that may be massively distributed over an area for remote tracking of human presence. Another application could be based on the fabrication of tiny physiological sensors implanted in humans, communicating with a single central unit in wireless body area networks (WBANs).

Technical requirements for human implant devices. There are several technical requirements that must be fulfilled by a human implantable device having sensing, computing and communication abilities.

In-vivo implants need to be as small as possible. It is impractical for an electronic device injected into the body to have a battery for supplying power. The large size of a battery compromises its utilisation, and a passive (also called battery-less or fully-autonomous) device may be preferred in practice.

Another requirement concerns the selectivity of the remote control of human biological functions. For example, remote control of a specific bio-molecular event inside a living cell has to be spatially-localised for other biological activities to remain unaffected in the cell. Bio-compatible material, hence, is required to be used for fabricating the implant device and its packaging, to avoid tissue reactions. The remote electronic control at, say, bio-molecular scale should not alter bio-molecular events occurring in the surrounding medium, leaving surrounding molecules relatively unaffected.

Technological feasibility of tiny devices implanted inside a living cell for the remote control of a wide range of biological functions or detection of presence of an antibody has